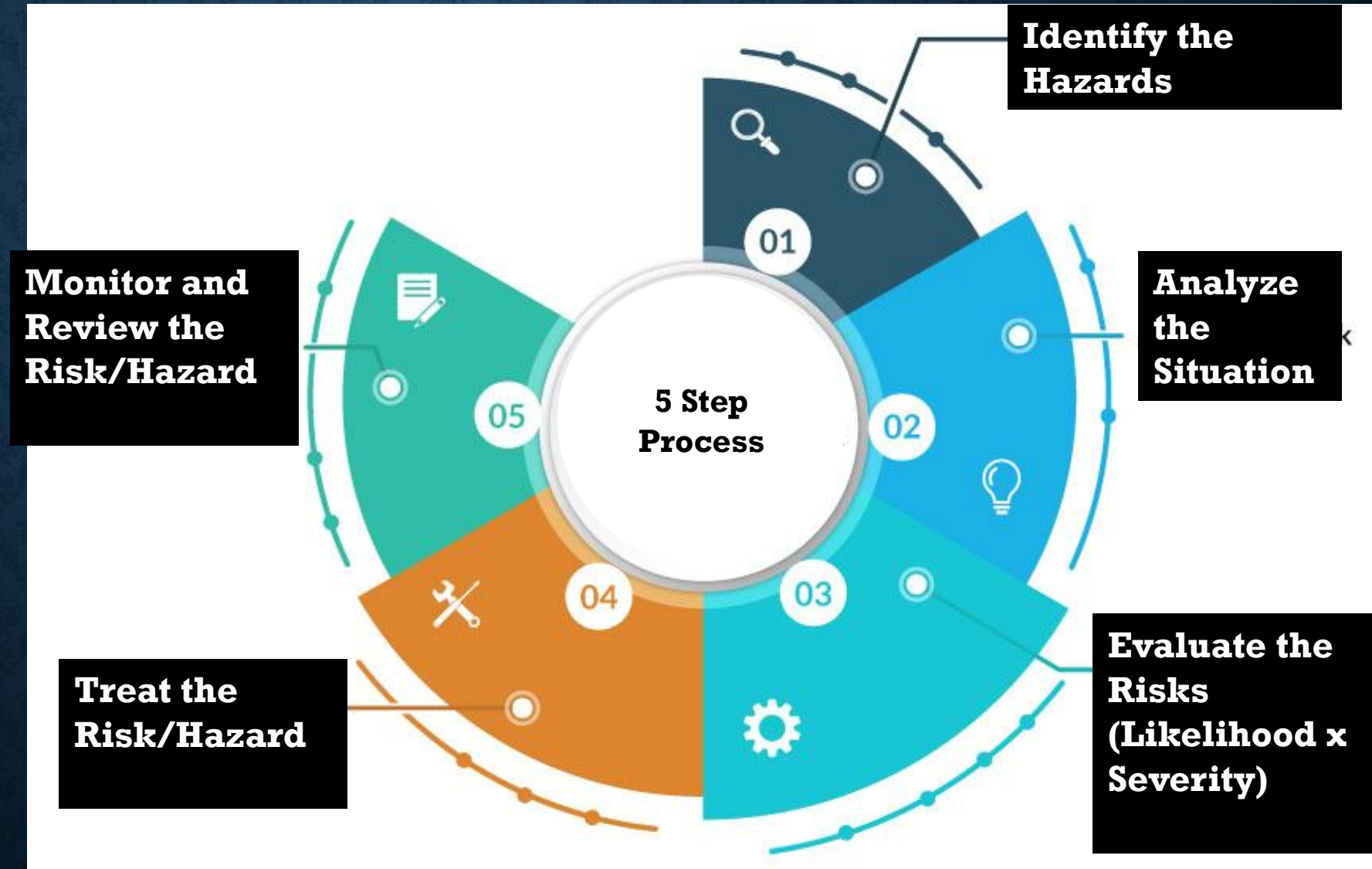


5-STEP PROCESS IS THE GLOBAL GOLD STANDARD FOR SAFETY MANAGEMENT.



STEP 1: IDENTIFY THE HAZARDS

- The goal here is to find anything with the "potential to cause harm."
- You don't just look with your eyes; you look at manuals, past accident reports, and talk to workers.

A large, heavy delivery truck reversing into a busy loading dock.

- The Hazard: The moving vehicle (Kinetic energy/Mechanical hazard).
- Secondary Hazards: Blind spots for the driver, loud engine noise (masking shouts), and slippery floor surfaces in the dock.

STEP 2: ANALYZE THE SITUATION

- Different people are at risk for different reasons.
- **The Driver:** Could hit a wall (whiplash).
- **Warehouse Staff:** Could be crushed between the truck and the dock.
- **Visitors/Pedestrians:** Might walk behind the truck, unaware of the blind spots.
- **How:** "Crushing injuries, bone fractures, or fatalities due to vehicle-pedestrian collision."

EVALUATE THE RISKS (LIKELIHOOD X SEVERITY)

- We assign a numerical value to prioritize what needs fixing first.
- **Likelihood (1-3):** If trucks arrive 20 times a day and there are no barriers, the likelihood is **3 (High)**.
- **Severity (1-3):** A truck crushing a person is a **3 (Fatality/Catastrophic)**.
- **Risk Score:** $3 \times 3 = 9$

3x3 Risk Matrix Template

		SEVERITY →		
		1	2	3
LIKELIHOOD ↓	1	LOW - 1 -	LOW - 2 -	MEDIUM - 3 -
	2	LOW - 2 -	MEDIUM - 4 -	HIGH - 6 -
	3	MEDIUM - 3 -	HIGH - 6 -	HIGH - 9 -

Smartsheet Inc. ©2025

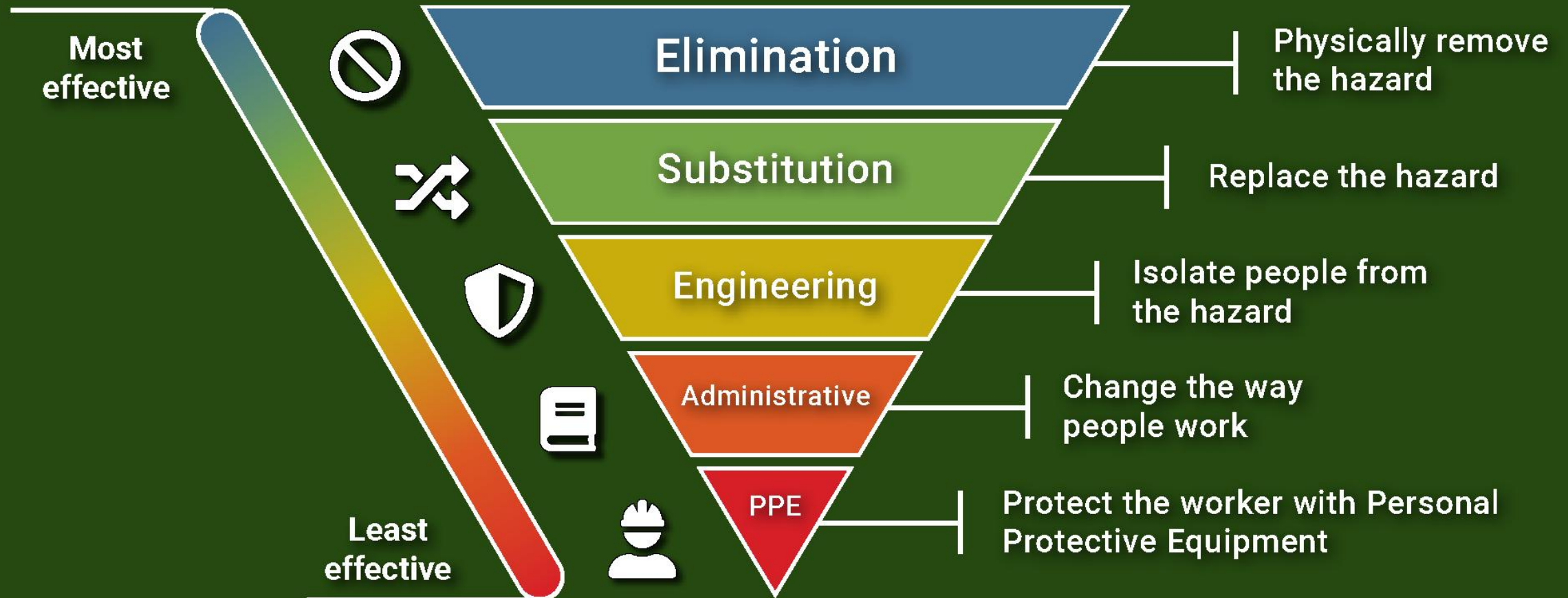
STEP 4: TREAT THE HAZARD OR RISK

The Hierarchy of Controls

Once a risk is identified, how do we fix it? We follow a strict order of effectiveness:

- Elimination: Physically remove the hazard.
- Substitution: Replace the hazard (e.g., use a less toxic chemical).
- Engineering Controls: Isolate people from the hazard (e.g., guardrails).
- Administrative Controls: Change the way people work (e.g., training, signs).
- PPE: Protect the worker with gear (e.g., gloves, helmets).

Hierarchy of Controls



Source: NIOSH

Dakota
SOFTWARE

Action	Responsible Party	Timeline
Hazard assessment	Safety Officers	1 Month
Step-by-Step Implementation Timeline		
Implement engineering controls	Maintenance Team	2-3 Months
Conduct training programs	HR & Safety Trainers	Ongoing
Provide PPE	Procurement Department	1 Month

HEALTH AND SAFETY AUDIT CHECKLIST

☒	PLANNING THE AUDIT
☐	Define the scope of the audit.
☐	Determine the frequency of the audits.
☐	Develop a detailed audit plan with timelines.
☐	Define the scope of the audit.
☒	RISK ASSESSMENT AND FOCUS AREAS
☐	Identify high-risk areas within the organization.
☐	Use the OSHA safety and health program audit tool to evaluate current risks.
☐	Prioritize areas for auditing based on historical data and known risks.
☒	CONDUCTING THE AUDIT
☐	Follow the audit checklist systematically.
☐	Document findings accurately and comprehensively.
☐	Use mobile and cloud solutions for real-time data collection and sharing.
☐	Evaluate policy effectiveness and compliance comprehensively.
☒	CONTINUOUS IMPROVEMENT AND FOLLOW-UP
☐	Schedule follow-up audits and status checks on corrective actions.
☐	Update the audit program based on new insights and regulatory changes.
☐	Engage in regular training updates and feedback sessions based on audit outcomes.
☐	Maintain open communication channels for ongoing safety discussions.
☒	DOCUMENTATION AND RECORDS
☐	Ensure access to previous audit reports.
☐	Gather safety training logs and incident reports.
☐	Review compliance documents and certifications.
☐	Check for up-to-date employee training records.
☒	TRAINING AND PREPARATION
☐	Conduct training sessions for all team members on audit procedures.
☐	Verify understanding of individual roles in the audit process.
☐	Review audit process steps and expected outcomes with the team.
☐	Ensure readiness of all team members to participate and cooperate during the audit.
☒	POST-AUDIT REVIEW
☐	Analyze audit findings to identify patterns and systemic issues.
☐	Prepare a detailed report including recommendations for improvement.
☐	Present findings to management and relevant stakeholders.
☐	Discuss corrective actions and set timelines for implementation.
☒	TOOLS AND RESOURCES
☐	List available health and safety audit software tools that can be used.
☐	Include templates and checklists for different auditing activities.
☐	Provide links to training resources and regulatory updates.

MONITORING AND REVIEWING CONTROLS

- Importance of regular monitoring to ensure controls remain effective.
- Methods - Inspections, audits, employee feedback, incident reports.
- Updating controls in response to changes in processes or regulations.

SCENARIO 5: THE LABORATORY

- **Situation:** Researchers are using a highly volatile solvent to clean glass slides. The work is currently being done on a standard open workbench, and several staff members have complained of headaches and dizziness by the end of the day.

SCENARIO 6: THE LOGISTICS WAREHOUSE

- Order pickers are required to lift 20kg boxes from floor-level pallets and place them onto a conveyor belt that is at waist height. This task is repeated roughly 150 times per shift. Two workers have reported chronic lower back pain.

SCENARIO 5: THE LABORATORY

Level of Control	Proposed Action
Elimination	Remove the solvent entirely and use a heat-cleaning process if applicable.
Substitution	Replace the volatile solvent with a water-based, non-toxic cleaning agent.
Engineering	Perform all slide cleaning inside a certified Fume Hood with active ventilation.
Administrative	Limit the time any one researcher spends on cleaning tasks (Job Rotation).
PPE	Provide high-grade respiratory masks and chemical-resistant gloves.

SCENARIO 6: THE LOGISTICS WAREHOUSE

Level of Control	Proposed Action
Elimination	Redesign the workflow so that pallets arrive on elevated platforms, removing the need to lift from the floor.
Substitution	Use a vacuum lifting tool or a robotic arm to handle the weight of the boxes.
Engineering	Install a scissor-lift table that keeps the pallet at waist height as layers are removed.
Administrative	Mandatory "Safe Lifting" training and frequent stretching breaks throughout the shift.
PPE	Provide back support belts (though these are often considered the least effective).